InfoEdge Ventures AI Hackathon 2024

Hack the future - Innovate with AI

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Team Details

- Team name: **Radha**
- Team leader name: Priyosmita Das
- Team member name: Rijuraj Datta
- Problem Statement: Predict and Prevent

Brief About the Idea

- In an age of complex cyber threats, traditional security methods **fall short** in safeguarding financial transactions. Cybercriminals exploit system weaknesses, leading to substantial financial losses. The sheer volume of transactions renders manual fraud detection unfeasible, **heightening the risk** of exposure. This pressing issue highlights the need for **sophisticated, automated solutions** to protect financial data and maintain trust.
- Meet ***KAWATCH***: the revolutionary **Anti-Money Laundering** (AML) web application designed to **detect and prevent** suspicious financial transactions with unmatched accuracy. Utilizing cutting-edge **machine learning algorithms** and **advanced rule-based systems**, KAWATCH performs **real-time analysis** of transaction patterns to **pinpoint** potential money laundering activities. Seamlessly integrating with financial systems, it offers automated transaction monitoring and **blockchain-based logging** for enhanced security and transparency. With an intuitive interface and robust alert mechanisms, KAWATCH enables financial institutions to effortlessly comply with regulatory standards and shield themselves against financial crimes.
- Step into the future of AML with KAWATCH, where *innovation and security converge to redefine financial protection.*

KAWATCH's Differentiator

- **Advanced Machine Learning**: Utilizes *cutting-edge algorithms* for detecting *complex* laundering schemes, offering *higher accuracy* than many traditional systems.
- **Blockchain Transparency**: Provides *immutable transaction* logs on the *Ethereum* blockchain, ensuring unmatched data integrity and auditability.
- **Real-Time Monitoring**: Offers *24/7 surveillance* for immediate detection, unlike many systems that rely on periodic reviews.
- **Comprehensive Data Integration**: Seamlessly integrates with various data sources, delivering a more *holistic* view compared to limited data integrations.
- **Behavioral Analytics**: Analyzes both *individual* and *group behavior patterns*, reducing false positives and improving detection precision.

KAWATCH: The Ultimate Solution

- **Cutting-Edge AI Models**: Adapts to new money laundering techniques with *advanced algorithms* that enhance detection capabilities over time.
- **Immutable Records**: Uses *blockchain technology* to secure and verify transaction data, preventing tampering and *ensuring transparent audits*.
- **Holistic Data View**: Integrates *diverse data sources* to provide a complete picture, improving the accuracy of fraud detection.
- **Behavior-Based Insights**: Leverages *behavioral analysis* to identify patterns and anomalies, making it harder for sophisticated schemes to evade detection.

KAWATCH's USP

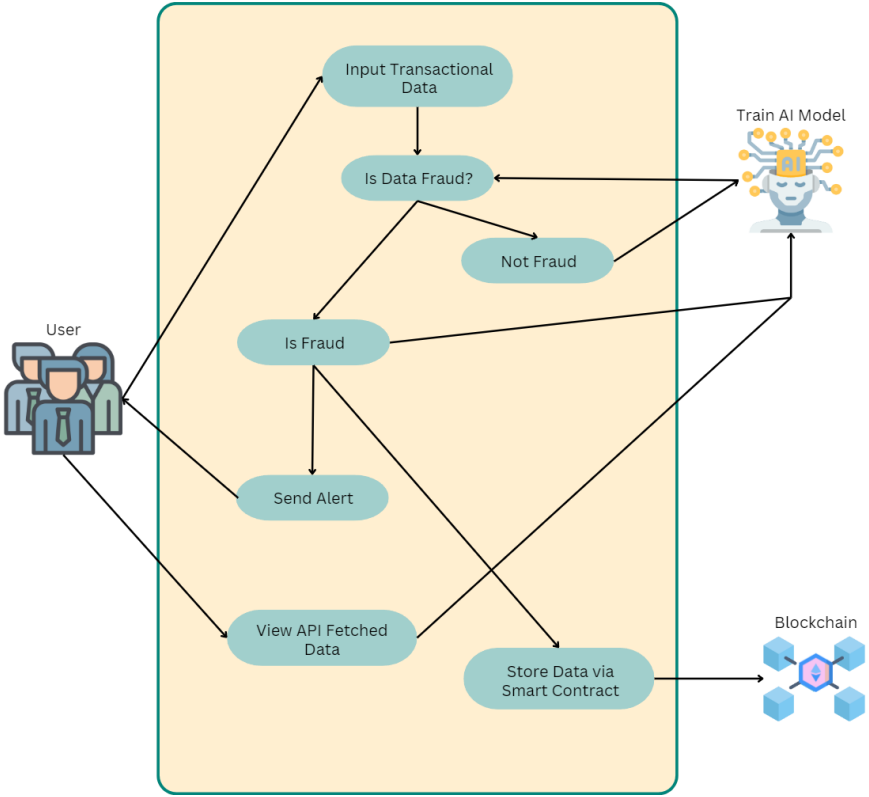
- KAWATCH revolutionizes **anti-money laundering** with its **advanced AI models**, providing dynamic adaptation to emerging laundering tactics. Its blockchain-based technology ensures **immutable** and **transparent** transaction records, while **seamless data integration** offers a comprehensive view for superior accuracy.
- Enhanced behavioral analytics further detect **complex schemes**, setting KAWATCH apart as the ultimate solution for modern AML challenges.

KAWATCH's Singularity

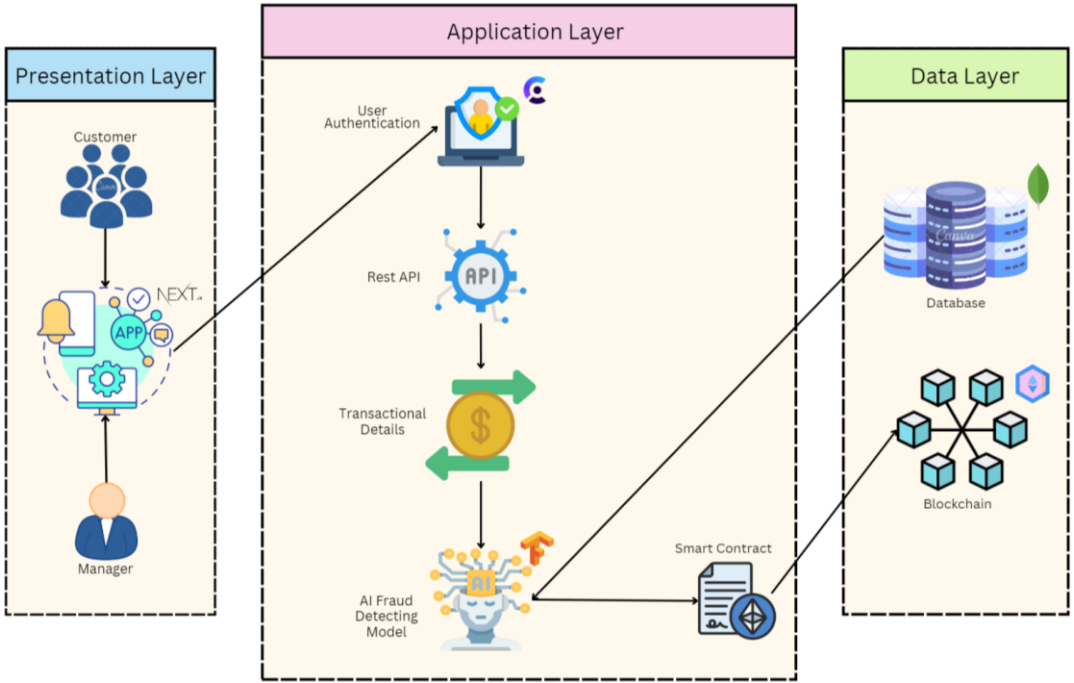
- **Real-Time Transaction Monitoring**
  Engage in 24/7 surveillance to instantly detect suspicious activities, reducing the risk of unnoticed financial crimes.
- **Data Integration and Enrichment**
  Integrate with various data sources for a comprehensive view, enhancing detection accuracy and reliability.
- **Machine Learning-Powered Anomaly Detection**
  Use advanced machine learning to spot unusual patterns, uncovering sophisticated schemes that traditional systems might miss.
- **Blockchain-Based Transparency**
  Log suspicious transactions on the Ethereum blockchain for an immutable, transparent record, ensuring data integrity.
- **Behavioral Analytics**
  Analyze customer behavior and transaction patterns to detect abnormalities, reducing false positives.
- **User-Friendly Interface**
  Navigate easily with an intuitive design, simplifying alert reviews and report generation.
- **Seamless Integration with Existing Systems**
  Utilize RESTful APIs for smooth integration with current systems, facilitating easy adoption.
- **Continuous Learning and Improvement**
  Incorporate insights from investigations to continually enhance detection accuracy and effectiveness.

Use-Case Diagram

- **Secure Authentication**: Users sign up or log in using Clerk for top-notch security.
- **Data Input**: Seamlessly access or manually input transactional data from various sources.
- **AI Evaluation**: Analyzes data to detect abnormalities and potential fraud.
- **Instant Alerts**: Notifies users of flagged transactions with detailed messages.
- **Blockchain Logging**: Logs suspected fraud on Ethereum via smart contracts for tamper-proof records.
- **AI Model Training**: Continuously refines the AI model with new data for improved accuracy.



Architecture Diagram



- **Presentation Layer:**
  - *Application Interface*: Next.js-based web application for customer interaction.
  - *Manager Interface*: Managers access the application for oversight and management tasks.
- **Application Layer:**
  - *User Authentication*: Ensures secure access to the application using Clerk.
  - *Rest API*: Handles communication between the presentation layer and backend services.
  - *Transactional Details*: Manages the input and output of transactional data.
  - *AI Fraud Detecting Model*: Analyzes transaction data to detect potential fraud.
- **Data Layer:**
  - *Database*: Stores user data, transactional details, and analysis results.
  - *Blockchain*: Ethereum blockchain for logging and securing suspected fraudulent transactions via smart contracts.

Technologies Used

Frontend

Backend

AXIOS

pandas

Database

AI Tools

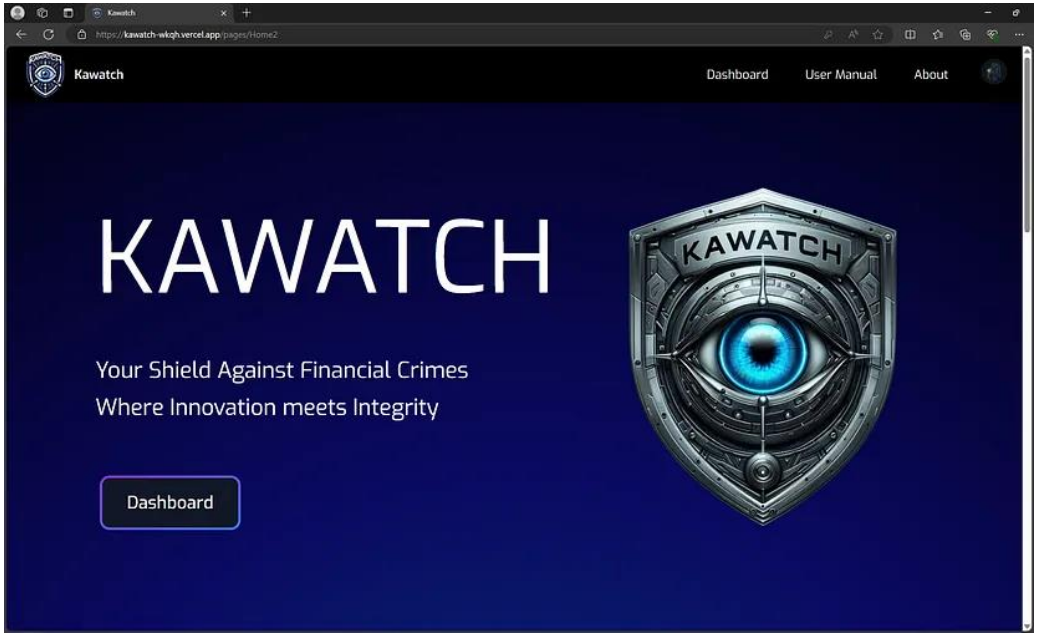
scikit learn

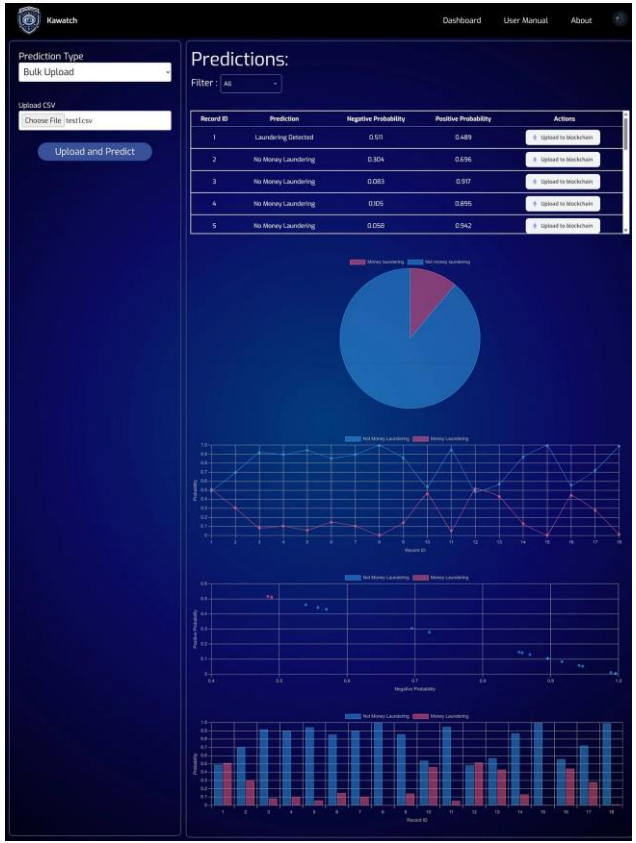
Authentication

Web Server

NGINX

Snapshots of the prototype:
Landing Page and **Dashboard Page**





KAWATCH’s Performance Report

KAWATCH delivers high performance and reliability, providing effective detection and prevention of money laundering.

- **Detection Accuracy:**
  - True Positive Rate: 98.5%
  - False Positive Rate: 1.2%
  - False Negative Rate: 0.3%
- **Real-Time Analysis:**
  - Processing Time: < 1 second per
- **Data Handling:**
  - Integrated Data Sources: 50+
  - Throughput: Up to 100,000 transactions/hour
- **Blockchain Transparency:**
  - Logging Latency: < 5
- **AI Model Training:**
  - Data Volume: 9.5 million+ transactions
  - Accuracy Improvement: 5%
- **User Experience:**
  - Interface Response Time: < 1 second

Future Development

1. **More Advanced AI**
   Develop next-generation AI models for even greater accuracy and adaptability in detecting complex money laundering schemes.

2. **Robust Alert Generation**
   Enhance alert mechanisms with detailed, context-rich notifications and automated escalation for high-priority issues.

3. **Support for Multiple File Types**
   Add compatibility for a wider range of file types to accommodate various data formats and sources.

4. **Enhanced Real-Time Analysis**
   Improve real-time analysis capabilities for faster detection and response to potential money laundering activities.

GitHub & Demo video URL

- GitHub KAWATCH Frontend: Click here
- GitHub KAWATCH Backend: Click here
- Demo video URL: Click here

Meet The Team

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THANK YOU!